Bryan Morales

Senior .NET Software Engineer | C#, .NET Core, Python, Angular, React, Azure, AWS

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Summary

Senior .NET Software Engineer with 10 years of professional experience developing tools, high-performance business applications, and workflow solutions across industries such as finance, healthcare, and banking. Led the development of a payment processing system leveraging multiple architectural patterns within a team of 13. Extensive experience in creating secure, compliant systems, including API security, cloud access control, and regulatory adherence. Greatest strength is always trying to remove any barriers out of the way on team's collaboration and productivity.

Skills

- C#, Python, JavaScript, TypeScript, SQL, T-SQL, Scala
- ASP.NET Core, .NET Core, Entity Framework Core, MediatR, Blazor, React, Next.js, Redux, Angular, RxJS
- Azure: Azure Functions, Azure DevOps, Azure Application Insights, Azure API Management, Azure Data Factory
- AWS: AWS Lambda, AWS Step Functions, AWS S3, AWS CloudFormation, AWS CloudWatch, AWS EKS
- SQL Server, MongoDB, Redis, DynamoDB, Azure Cosmos DB, ElasticSearch
- Terraform, Docker, Kubernetes, Jenkins, Azure DevOps (ADO), GitHub Actions, AWS CodePipeline
- NUnit, xUnit, Cypress, Selenium WebDriver, Postman, JMeter, SonarQube
- Agile Methodology, Scrum, JIRA, Confluence, Trello

Work Experience

Senior Software Engineer, Team Lead
TapCheck
Mar 2022 – Oct 2024 | Plano, TX

- Architected and developed a fault-tolerant microservices-based payment processing platform using **C#**, **.NET Core 6**, **React**, **Next.js**, **SQL Server**, and **RabbitMQ**, incorporating **Saga**, and **Event Sourcing** patterns to ensure **PCI DSS** compliance, reliable transaction processing, and fault tolerance for payroll deductions and financial workflows.
- Refactored microfrontend architecture with **React** and **Next.js**, improving rendering performance by 20% using **React Hooks**, memoization, and **Next.js ISR/SSR** strategies for high-traffic payroll dashboards and transaction workflows.
- Developed and maintained mission-critical backend services, **RESTful API**s, and **GraphQL** endpoints using **.NET Core** and **Node.js**, enabling the processing of millions of ACH transactions daily and enhancing system throughput by 25%.
- Enhanced backend API security by implementing **OAuth 2.0**, **JWT** authentication, IP whitelisting, and encrypted token storage, ensuring compliance with financial regulations and preventing unauthorized access.
- Hardened search performance by integrating **Elasticsearch**, optimizing indexing and retrieval logic to reduce query latency by 50%, and alleviating **SQL Server load**.
- Engineered a real-time notification service using **SignalR** (**C#)** and **WebSockets** (**Node.js**) to provide instant payroll status updates, transaction alerts, and employer-employee communication, improving user engagement and reducing support inquiries.
- Improved database performance by implementing advanced indexing, partitioning, and caching strategies in **SQL Server** and **Redis**, optimizing query execution times by 30% under peak transaction loads.
- Forged interactive analytics dashboards with **Power BI** and **Tableau**, delivering real-time financial insights, payroll trends, and payment analytics, improving business decision-making efficiency by 30%.
- Created schedulable customized data extraction process using **Azure Functions**, API calls and **Service Bus Queues** to generate formatted reports and securely deliver them via **FTP**, enhancing data accessibility and automation efficiency.
- Automated performance monitoring and alerting with **Azure Application Insights** and **Kusto** queries, reducing troubleshooting time by 40% and proactively detecting system anomalies.
- Implemented **CI/CD** best practices, integrating **Azure DevOps**, **Terraform**, and feature flag management, achieving 80%+ automated test coverage using **NUnit** and **Cypress**, reducing regression testing time by 50%.

- Led API integrations with third-party logistics via **Azure API Management**, incorporating rate limiting, **API versioning**, and caching, ensuring high availability and seamless scalability for external financial systems.
- Collaborated with product managers and stakeholders to align technical solutions with business goals, delivering features that enhanced customer satisfaction and operational efficiency.

Senior Software Engineer
Hallmark Health Care Solutions
Sep 2018 – Feb 2022 | Charlestown, MA

- Directed the migration of a legacy **ASP.NET WebForms** physician management platform backend to a modern **ASP.NET Core**-based microservices architecture, leveraging **AWS Lambda** and **Step Functions** to enhance scalability and reduce cloud costs by 25%.
- Formulated a scalable **WebAPI** template using **Vertical Slice architecture** and the **MediatR** library, incorporating **CQRS** and **Domain-Driven Design** (**DDD**) principles to define domain models and bounded contexts, improving modularity and reducing API response times by 15%.
- Optimized high-demand healthcare workflows by migrating critical API endpoints to **gRPC**, reducing network latency by 25% and improving inter-service communication efficiency.
- Initiated interactive **Blazor** components for patient dashboards and claims processing, integrating real-time data visualization, improving UI responsiveness by 35%, and enhancing provider-patient engagement.
- Enhanced database performance using **Entity Framework Core** with advanced **LINQ** queries, lazy/eager loading strategies, and seamless database migrations, reducing query times by 30% and ensuring reliability during schema updates.
- Strengthened healthcare analytics pipelines by building **Databricks** notebooks (**PySpark** & **Scala**) to process large-scale genomic and clinical datasets from **Azure Data Lake**, improving data processing efficiency by 30%.
- Automated end-to-end testing using **Selenium WebDriver**, **Postman**, and **JMeter**, achieving high reliability in patient portals and claims workflows and ensuring robust API stability under heavy transaction volumes.
- Designed a multi-stage **API Gateway** to manage high-volume healthcare transactions, incorporating **VPC Endpoints**, by developing **.NET** & **Node.js Lambdas** to handle authorization, and proxying of requests to the services hosted in **EKS** Cluster.
- Streamlined backend workflows by transitioning legacy **ETL** processes to microservices and serverless architectures with **AKS**, **EKS**, and **AWS Lambdas**, reducing infrastructure costs by 70% and improving reliability for claims and records processing.
- Executed a multi-threaded **Kafka** consumer service for real-time processing of high-volume transactions, such as claims and appointment confirmations, with low latency and high reliability.
- Improved system observability by configuring **AWS CloudWatch**, **Grafana** Dashboards, and **Splunk**, reducing issue resolution times by 40% and enhancing system reliability.

Software Engineer
Green Dot
Jul 2014 – Sep 2018 | West Chester, OH

- Formed and documented **RESTful APIs** with well-defined **JSON** schemas and implemented **API** versioning to support evolving business requirements for ACH transfers, reducing development time by 25%.
- Performed **ASP.NET Web Forms** validations using **.NET** Validation Controls and **Angular 4** for client-side validation, leveraging **Telerik** controls for interactive data visualization to enhance user experience and ensure data accuracy.
- Executed the integration and testing of more than 10 distinct Web Services coupled with **WCF** services, ensuring accurate transaction handling for ACH transfers validated through **SoapUI**, which improved overall system efficiency.
- Architected a data access layer using **ADO.NET** to interact with **Amazon RDS**, crafting optimized **T-SQL** stored procedures to streamline transaction tracking and reporting, reducing operational errors by 20%.

Education

- The University of Texas at Dallas (2009 – 2014) | Bachelor of Science in Computer Science